Microprice as a Martingale Corrector

A Unified Markov–Operator Framework for Fair Value and Value-Function Extensions

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Abstract. Stoikov (2018) introduced the *microprice* as a horizon-free limit of conditional expectations of future prices, computed from Level-I order book data via an absorbing Markov chain. We reframe that construction as an operator statement about Markov additive processes at mid-price move times.

At move times the mid-price increment has a state-dependent conditional mean. We show that Stoikov's adjustment term is exactly the *Poisson corrector* (a Markov potential) on the post-move chain: the unique centered solution of a Poisson equation. Adding this corrector to the mid-price produces a martingale. The same Poisson equation yields martingale correctors for *any* additive order-book functional.

Many "fair value" objects can be expressed as value functions V solving $V = u + \Gamma M V$ for a Markov operator M and a continuation/discount operator Γ with $\|\Gamma M\| < 1$. In this regime the solution is a resolvent $V = (I - \Gamma M)^{-1} u$ with a convergent Neumann series. The microprice corresponds to the boundary case $\Gamma = I$: the resolvent becomes singular and well-posedness is recovered by centering. We provide clean existence/uniqueness conditions, operator-norm bounds for the absorbing-chain fundamental matrix, and truncation-error control for the main series solutions.

Keywords: microprice, martingale corrector, Poisson equation, Markov additive process, Markov potential, Neumann series, resolvent, absorbing Markov chain

Contents

1 Introduction **4**
- 1.1 Two guiding ideas . . . 4
- 1.2 Contributions . . . 4
- 1.3 Related work and terminology . . . 4
- 1.4 Roadmap . . . 5

2 Setup and notation **5**
- 2.1 Prices and book features . . . 5
- 2.2 Event time vs. move time . . . 6
- 2.3 State discretization . . . 6
- 2.4 Standing assumptions . . . 6

3 Microprice mechanics: from absorbing event time to a Poisson equation **7**
- 3.1 Transition decomposition . . . 7
- 3.2 The fundamental matrix of the no-move kernel . . . 8
- 3.3 Post-move chain and first adjustment . . . 8
- 3.4 The microprice adjustment as a Poisson equation . . . 9
- 3.5 Microprice as a martingale corrector . . . 10

4 Microprice as a general martingale constructor **11**
- 4.1 Doob decomposition for state observables . . . 11
- 4.2 Poisson correctors for additive functionals . . . 12
- 4.3 Operator notation: potentials, projections, and generalized "micro-" transforms 12
- 4.4 Vector-valued functionals . . . 13

5 Value functions and Neumann-series resolvents **13**
- 5.1 Neumann-series value functions . . . 14
- 5.2 Examples of continuation operators . . . 14
- 5.3 Microprice as a centered boundary case . . . 15

6 Discussion and limitations **15**

7 Conclusion **15**

A Proofs **17**

1 Introduction

Stoikov's microprice [1] starts from a practical microstructure question: given the current state of the limit order book, what is a fair estimate of the asset's value? The classical mid-price

$$p_t := \frac{b_t + a_t}{2}$$

ignores information in queue imbalance and related order-flow imbalance signals [4, 2]. A fair value should incorporate how the current book state predicts future mid-price moves.

1.1 Two guiding ideas

Our goal is to make explicit two ideas that are present but implicit in the original construction.

At *move times* (times when the mid-price changes), the mid-price forms a Markov additive process: the state evolves as a finite Markov chain and the price increment has a state-dependent conditional mean. Stoikov's "microprice adjustment" is the Poisson corrector that removes this predictable component, turning price-plus-corrector into a martingale. This viewpoint immediately generalizes: the same Poisson equation produces martingale correctors for any additive order-book functional.

Once the relevant Markov operator is identified, a broad class of "fair value" objects are value functions V solving a linear fixed-point equation of the form

$$V = u + \Gamma M V,$$

where M is a Markov operator and Γ encodes discounting/continuation. When $\|\Gamma M\| < 1$, the inverse $(I - \Gamma M)^{-1}$ exists and the solution is given by a convergent Neumann series. The microprice corresponds to the boundary case $\Gamma = I$, where the resolvent becomes singular and well-posedness is recovered by centering.

1.2 Contributions

This note makes three contributions: (i) it identifies Stoikov's microprice adjustment as the centered solution of a Poisson equation on the post-move chain; (ii) it generalizes the construction to a martingale corrector for any additive order-book functional with state-dependent drift; (iii) it embeds both the absorbing fundamental matrix $(I - Q)^{-1}$ and the microprice operator in a single resolvent family $V = (I - \Gamma M)^{-1} u$, with the microprice recovered as the centered limit when the discount vanishes ($\Gamma \uparrow I$).

1.3 Related work and terminology

This note is primarily mathematical, but it sits at the intersection of two literatures.

On the microstructure side, Level-I (best bid/ask) order-book features such as spread and imbalance are widely used to predict short-horizon price changes; see the survey [4] and, for empirical evidence that order-flow (and hence queue) imbalances drive price changes, [2]. Stoikov's microprice [1] formalizes a horizon-free "fair value" using an absorbing Markov chain built from these features. Markovian order-book models that explicitly embed price changes as an event-driven Markov system appear, for example, in [3].

On the Markov-chain side, the absorbing-chain fundamental matrix $N = (I - Q)^{-1}$ and its expected-visit interpretation are classical [5]. For ergodic chains, the matrix $(I - B + \Pi)^{-1}$

(with $\Pi = \mathbf{1}\pi^T$) and the related group inverse $(I - B)^{\#}$ play a central role in potential theory and first-passage identities [6]. The Poisson equation and its connection to martingale decompositions for additive functionals are standard tools in Markov chain theory and limit theorems; see, e.g., [10, 11]. Finally, the fixed-point/resolvent viewpoint in Section 5 is the familiar value-function perspective from Markov decision processes and dynamic programming [13].

1.4 Roadmap

Section 2 fixes the event-time and move-time models and notation. Section 3 derives the post-move chain, the first adjustment, and the Poisson equation behind Stoikov's microprice. Section 4 elevates this to a general martingale-constructor theorem for additive order-book functionals. Section 5 introduces Neumann-series value functions and explains the microprice operator as a centered boundary case. Section 6 discusses modelling assumptions and practical conditioning issues. Proofs are collected in Appendix A.

2 Setup and notation

We work with Level-I order book data [4, 3]. At each event time t we observe the best bid b_t, best ask a_t, and their queue sizes B_t and A_t.

2.1 Prices and book features

We use the following derived quantities:

$$\begin{aligned}
\text{Mid-price:} \quad & p_t = (b_t + a_t)/2, \\
\text{Spread:} \quad & s_t = a_t - b_t, \\
\text{Imbalance:} \quad & i_t = \frac{B_t - A_t}{B_t + A_t} \in [-1, 1], \\
\text{Size-weighted mid (instantaneous “microprice” proxy):} \quad & m_t = \frac{a_t B_t + b_t A_t}{B_t + A_t} = p_t + i_t \cdot \frac{s_t}{2}.
\end{aligned} \tag{1}$$

Mapping to Stoikov (2018) notation. Stoikov [1] uses an imbalance $I_{\mathrm{S}} \in [0, 1]$ and the *half-spread* $S = \frac{1}{2}(a_t - b_t)$. Our imbalance $i_t \in [-1, 1]$ and *full spread* $s_t = a_t - b_t$ are related by

$$i_t = 2I_{\mathrm{S}} - 1, \qquad s_t = 2S.$$

With this identification, the size-weighted mid identity agrees exactly with Stoikov's weighted mid:

$$m_t = p_t + i_t \frac{s_t}{2} = p_t + (2I_{\mathrm{S}} - 1)S.$$

The object of interest is not m_t itself, but rather a *fair value* obtained by conditioning on the book state and looking forward to future price moves.

2.2 Event time vs. move time

There are two natural clocks. Event time counts every book update (queue change, trade, etc.); the discretized state evolves as a Markov chain at this clock. Move time counts only events that change the mid-price, and this embedded chain is the right object for the microprice/martingale-corrector viewpoint.

Let $\tau_0 = 0$ and let

$$\tau_{k+1} := \inf\{t > \tau_k : p_t \neq p_{t-}\} \tag{2}$$

be the $(k+1)$-st mid-price move time. Define the move-time price and state

$$p_k := p_{\tau_k}, \qquad X_k := X_{\tau_k}.$$

We write $\mathcal{F}_k := \sigma(\,(p_j, X_j)_{0\le j\le k}\,)$ for the *move-time filtration*. All "fair value" objects in this paper are conditional expectations given the information available at time k, i.e. conditional on $\mathcal{F}_k$ (not merely on X_k).

Because the mid-price is additive across moves, it is natural to forecast *increments* relative to the current price level. Under the Markov-additive structure formalized in Assumption 3, the conditional law of future price increments depends on the book state X_k but not on the absolute price level p_k. Consequently, many conditional expectations take the form "current value + state-dependent correction".

2.3 State discretization

We discretize continuous features into buckets. For the single-instrument setting, a common choice is

$$X_t = (\text{spread bucket}, \text{imbalance bucket}),$$

with m spread buckets and n imbalance buckets, giving $L = mn$ states. Nothing in the operator algebra depends on this particular choice; any finite Markov state can be used.

2.4 Standing assumptions

Assumption 1 (Markov property)**.** The discretized state X_t is Markov at event times.

Assumption 2 (Bounded increments)**.** One-step mid-price changes satisfy $|\Delta p_t| \le D$ for some constant D.

Assumption 3 (Markov additive structure / translation invariance)**.** At move times, the pair (X_k, p_k) is a *Markov additive process* in the following sense: conditional on $\mathcal{F}_k$, the law of $(X_{k+1}, \Delta p_{k+1})$ depends on the past only through X_k, and the conditional distribution of the increment $\Delta p_{k+1} := p_{k+1} - p_k$ depends on X_k but not on the absolute price level p_k. Equivalently, there exists a function $g_1 : \mathcal{X} \to \mathbb{R}$ such that $\mathbb{E}[\Delta p_{k+1} \mid \mathcal{F}_k] = g_1(X_k)$.

Assumption 4 (Ergodicity at move times)**.** The post-move chain ($B := (I - Q)^{-1}T$, defined in Section 3) is irreducible and aperiodic, admitting a unique stationary distribution.

3 Microprice mechanics: from absorbing event time to a Poisson equation

This section makes explicit the mechanics behind Stoikov's microprice computation [1]. We first decompose the event-time kernel into no-move and move pieces (Q, T), then use the fundamental matrix $(I-Q)^{-1}$ to compute the embedded post-move chain B and the one-move drift G_1, and finally recognize the infinite-horizon adjustment as the centered solution to a Poisson equation on B.

3.1 Transition decomposition

Partition event-time transitions according to whether the mid-price moves. Let Q_{xy} denote the probability of transitioning from state x to y without a price move, and T_{xy} the probability with a move. The full event-time transition matrix is

$$P := Q + T,$$

and is row-stochastic.

Let

$$p_{\text{move}}(x) := \sum_y T_{xy} = 1 - \sum_y Q_{xy}$$

be the probability the next event is a mid-price move.

For price changes, we allow the (conditional) move size to depend on the *move transition*. Let

$$\Delta^p_{xy} := \mathbb{E}[\Delta p \,|\, X_{n-1} = x,\ X_n = y,\ \text{the step } (x \to y) \text{ is a price move}]$$

denote the conditional mean mid-price change associated with a move transition $x \to y$. Collect these means into the matrix $\Delta^p \in \mathbb{R}^{L\times L}$. Then the unconditional one-step drift at event time (including zero change on no-move events) is

$$\bar{r}_x := \mathbb{E}[\Delta p_n \mid X_{n-1} = x] = \sum_y T_{xy}\,\Delta^p_{xy}, \qquad \text{i.e.} \qquad \bar{r} = (T \odot \Delta^p)\,\mathbf{1}, \tag{3}$$

where $\odot$ denotes entrywise (Hadamard) product and $\mathbf{1}$ is the all-ones column vector.

A common special case is that the move size distribution depends only on the *pre-move* state x and takes values $\{\delta_j\}_j$ (e.g. $\pm$ one tick). Writing $R_{xj} := \mathbb{P}(\Delta p = \delta_j \mid X_{n-1} = x,\ \text{move})$ and $\boldsymbol{\delta}$ for the vector with entries δ_j, the conditional mean move size is

$$r^{\text{move}}_x := \sum_j R_{xj}\,\delta_j, \qquad \text{i.e.} \qquad r^{\text{move}} = R\,\boldsymbol{\delta}.$$

In this factorized setting $\Delta^p_{xy} = r^{\text{move}}_x$ does not depend on y and (3) reduces to

$$\bar{r}_x = p_{\text{move}}(x)\, r^{\text{move}}_x, \qquad \text{i.e.} \qquad \bar{r} = D_{p_{\text{move}}}\, r^{\text{move}},$$

where $D_{p_{\text{move}}} := \operatorname{diag}(p_{\text{move}}(x))$.

3.2 The fundamental matrix of the no-move kernel

Between consecutive price moves, the book state evolves according to the no-move kernel Q. Understanding this transient dynamics is essential: the fundamental matrix $(I-Q)^{-1}$ aggregates expected visits before the next move, and its invertibility guarantees that moves eventually arrive from every state.

The no-move matrix Q is substochastic: each row sums to at most 1. When moves occur with uniformly positive probability, the spectral radius of Q is strictly less than one, and the fundamental matrix exists.

Definition 3.1. A matrix $Q \in \mathbb{R}^{L\times L}$ is *substochastic* if $Q_{xy} \geq 0$ for all x, y and $\sum_y Q_{xy} \leq 1$ for all x.

Lemma 3.2 (Spectral radius bound)**.** *Let Q be substochastic with $\sum_y Q_{xy} \leq 1-\varepsilon$ for all x. Then $\rho(Q) \leq 1-\varepsilon < 1$.*

Proposition 3.3 (Fundamental matrix)**.** *Under the conditions of Lemma 3.2, the matrix $I-Q$ is invertible with*

$$N := (I-Q)^{-1} = \sum_{k=0}^{\infty} Q^k, \qquad \|N\|_\infty \leq \frac{1}{\varepsilon}.$$

The entries N_{xy} equal the expected number of visits to y, starting from x, before the first price move [5].

Remark 3.4 (Sufficient versus necessary conditions for $I-Q$ invertibility). Lemma 3.2 assumes a uniform bound $\sum_y Q_{xy} \leq 1-\varepsilon$, i.e. a state-independent lower bound on move probability. This is a convenient *sufficient* condition yielding the clean norm bound $\|N\|_\infty \leq 1/\varepsilon$. For the microprice algebra itself, the essential requirement is the *transience* of the no-move subchain, equivalently $\rho(Q) < 1$ (eventual arrival of a move almost surely from every state).

Lemma 3.5 (Non-negativity)**.** *The fundamental matrix N has non-negative entries, with $N_{xx} \geq 1$ for all x.*

3.3 Post-move chain and first adjustment

Definition 3.6. The *first adjustment* is the expected cumulative price change up to the first move:

$$G_1 := (I-Q)^{-1}\bar{r} = N\,\bar{r}.$$

In the factorized setting $\bar{r} = D_{p_{\text{move}}} r^{\text{move}}$ (see (3)), this reads $G_1 = N D_{p_{\text{move}}} r^{\text{move}}$. The *post-move chain* is the transition matrix observed only at move times:

$$B := (I-Q)^{-1}T = NT.$$

Lemma 3.7 (Embedded-chain interpretation)**.** *Let $(X_n)_{n\geq 0}$ be the event-time Markov chain with transition matrix $P = Q+T$, and let*

$$\tau := \inf\{n \geq 1 : \textit{the step } (X_{n-1} \to X_n) \textit{ is a price move}\}.$$

Then $B_{xy} = \mathbb{P}_x(X_\tau = y)$ is the distribution of the post-move state, and the first adjustment is

$$G_1(x) = \mathbb{E}_x[\Delta p_\tau],$$

the expected mid-price change at the next move. Equivalently, G_1 is the unique solution of the first-step system

$$G_1 = \bar{r} + QG_1. \tag{4}$$

Lemma 3.8 (Move-time chain is Markov with kernel B)**.** *Let $(X_n)_{n\geq 0}$ be the event-time Markov chain with transition matrix $P = Q + T$, and let $(\tau_k)_{k\geq 0}$ be the successive move times as in (2). Then the sampled process $(X_{\tau_k})_{k\geq 0}$ is a Markov chain on $\mathcal{X}$ with transition kernel B:*

$$\mathbb{P}\big(X_{\tau_{k+1}} = y \mid \mathcal{F}_{\tau_k}\big) = B_{X_{\tau_k},y} \qquad a.s.$$

In particular, conditional on $X_{\tau_k} = x$, the law of the next post-move state is $B_{x\cdot}$.

Lemma 3.9. *If $Q + T$ is row-stochastic and $\rho(Q) < 1$, then B is row-stochastic.*

3.4 The microprice adjustment as a Poisson equation

Fix a move-time index k and recall the move-time filtration $\mathcal{F}_k = \sigma((p_j, X_j)_{0\leq j\leq k})$. For $n \geq 0$, define the *n-move fair value* (the best mean-square predictor of the future mid-price at horizon n moves) by

$$p_k^{(n)} := \mathbb{E}[p_{k+n} \mid \mathcal{F}_k]\,.$$

Because p_k is $\mathcal{F}_k$-measurable and the mid-price is additive across moves, we can write

$$p_{k+n} = p_k + \sum_{\ell=1}^{n} \Delta p_{k+\ell}, \qquad \Delta p_{k+\ell} := p_{k+\ell} - p_{k+\ell-1}.$$

Under the Markov-additive/translation-invariance Assumption 3, the conditional law of future increments given $\mathcal{F}_k$ depends on the past only through X_k. Hence there exists a deterministic function $G^{(n)} : \mathcal{X} \to \mathbb{R}$ such that

$$p_k^{(n)} \;=\; p_k \;+\; G^{(n)}(X_k), \qquad \text{where} \qquad G^{(n)}(x) := \mathbb{E}[p_{k+n} - p_k \,|\, X_k = x]\,. \tag{5}$$

By time-homogeneity of the move-time dynamics, $G^{(n)}$ does not depend on k.

Remark 3.10 (Conditioning on X_k versus conditioning on $\mathcal{F}_k$). If one conditions only on X_k (and not on the current price level), then $\mathbb{E}[p_{k+n} \mid X_k] = \mathbb{E}[p_k \mid X_k] + G^{(n)}(X_k)$. In trading applications, p_k is observed, so the relevant fair value is (5).

When the limit exists, the *microprice* (horizon-free fair value at move times) is

$$p_k^{\text{micro}} := \lim_{n\to\infty} p_k^{(n)} = p_k + G^*(X_k), \qquad \text{where} \qquad G^* := \lim_{n\to\infty} G^{(n)}.$$

By the tower property and the Markov property at move times, the adjustments satisfy the recursion

$$G^{(0)} = 0, \qquad G^{(n)} = G_1 + BG^{(n-1)}.$$

Unrolling gives

$$G^{(n)} = \sum_{\ell=0}^{n-1} B^\ell G_1.$$

When it converges, the infinite-horizon adjustment is

$$G^* := \lim_{n\to\infty} G^{(n)} = \sum_{\ell=0}^{\infty} B^\ell G_1. \tag{6}$$

Since B is stochastic (hence has eigenvalue 1), the convergence of (6) requires a *centering* condition.

Definition 3.11 (Centering)**.** Let π be the stationary distribution of B. We say G_1 is *centered* if $\pi^T G_1 = 0$.

Remark 3.12 (Equivalence with Stoikov's $B^* G_1 = 0$ condition). For a finite ergodic chain, $B^k \to \mathbf{1}\pi^T$ as $k \to \infty$, so Stoikov's limit kernel satisfies $B^* = \mathbf{1}\pi^T$. Hence $B^* G_1 = 0$ is equivalent to $\pi^T G_1 = 0$.

Theorem 3.13 (Microprice convergence)**.** *Let B be an ergodic Markov kernel on a finite state space with stationary distribution π. Let*

$$\lambda_* := \max\{|\lambda| : \lambda \in \sigma(B),\ \lambda \neq 1\} \in [0, 1)$$

be the second-largest eigenvalue modulus (SLEM). See, e.g., [8, 9] for background on the spectral picture of finite ergodic chains. If G_1 is centered, i.e. $\pi^T G_1 = 0$, then:

1. *The series (6) converges absolutely to*

$$G^* = (I - B + \mathbf{1}\pi^T)^{-1} G_1;$$

   see [5, 6]. In particular, G^* is centered: $\pi^T G^* = 0$.

2. *G^* is the unique solution of the centered Poisson equation*

$$(I - B)G^* = G_1, \qquad \pi^T G^* = 0. \tag{7}$$

3. *For any $\rho \in (\lambda_*, 1)$ there exists $C_\rho < \infty$ such that the truncation error after n terms satisfies*

$$\left\| G^* - \sum_{k=0}^{n-1} B^k G_1 \right\|_\infty \leq \frac{C_\rho \rho^n}{1-\rho} \|G_1\|_\infty.$$

3.5 Microprice as a martingale corrector

At move times, (X_k, p_k) is a Markov additive process:

$$\mathbb{P}(X_{k+1} = y \mid X_k = x) = B_{xy}, \qquad \mathbb{E}[\Delta p_{k+1} \mid X_k = x] = G_1(x),$$

where $\Delta p_{k+1} := p_{k+1} - p_k$.

Theorem 3.14 (Microprice martingale)**.** *Assume the centered Poisson equation* (7) *has solution* G^* *(equivalently,* $\pi^T G_1 = 0$*). Then the move-time process*

$$\mathcal{P}_k := p_k^{\text{micro}} := p_k + G^*(X_k) \tag{8}$$

is a martingale with respect to the move-time filtration.

Remark 3.15 (Trending markets and discounted surrogates). If $\mu := \pi^T G_1 \neq 0$ (nonzero long-run drift per move), the undiscounted series (6) diverges linearly. One still has a martingale decomposition:

$$p_k - k\mu + \tilde{G}(X_k) \ \text{ is a martingale,}$$

where $\tilde{G}$ solves the centered Poisson equation $(I-B)\tilde{G} = G_1 - \mu\mathbf{1}$. In trending/regime-shifting markets, one should either remove the stationary drift (as in Theorem 4.1) or work with a well-posed value function satisfying a Neumann-series condition (Section 5).

Proposition 3.16 (First adjustment bounds)**.** *If the mid-price move size is uniformly bounded by* $|\Delta p| \leq D$*, then* $\|G_1\|_\infty \leq D$*.*

If one instead bounds via the operator norm, $\|G_1\|_\infty \leq \|N\|_\infty \|\bar{r}\|_\infty$*, and imposes an absorption floor* $p_{\text{move}}(x) \geq \varepsilon$ *so that* $\|N\|_\infty \leq 1/\varepsilon$*, then* $\|G_1\|_\infty \leq \|\bar{r}\|_\infty/\varepsilon$*.*

Remark 3.17 (Behavior off the centered subspace). For a general $v \in \mathbb{R}^L$, decompose $v = (\pi^T v)\mathbf{1} + (v - (\pi^T v)\mathbf{1})$ into its constant and centered parts. Since $B\mathbf{1} = \mathbf{1}$, one has the explicit resolvent identity

$$(I - \alpha B)^{-1} v = \frac{\pi^T v}{1-\alpha}\,\mathbf{1} \ + \ (I - \alpha B)^{-1}\big(v - (\pi^T v)\mathbf{1}\big).$$

Thus the only obstruction at $\alpha \uparrow 1$ is the constant component: the centered component has a finite limit given by Proposition 5.2.

4 Microprice as a general martingale constructor

The microprice construction in Section 3 used only two properties of the mid-price: additivity across moves and a state-dependent conditional mean increment. These two properties suffice for a general martingale construction. Any additive functional driven by a Markov state admits a predictable drift term plus a martingale term, and the martingale term is obtained by solving a Poisson equation.

4.1 Doob decomposition for state observables

Let $(X_k)_{k\geq 0}$ be a Markov chain with transition matrix B. For any bounded function $f : \mathcal{X} \to \mathbb{R}$, the process $f(X_k)$ admits the decomposition

$$f(X_k) = f(X_0) + M_k + A_k, \tag{9}$$

where

$$A_{k+1} - A_k = (Bf - f)(X_k), \qquad M_{k+1} - M_k = f(X_{k+1}) - Bf(X_k),$$

and (M_k) is a martingale. Equation (9) is the discrete-time Doob decomposition; see, e.g., [12].

4.2 Poisson correctors for additive functionals

Now let $(Y_k)_{k\geq 0}$ be an *additive* functional driven by the chain:

$$Y_{k+1} = Y_k + \xi_{k+1}, \qquad \bar{\xi}(x) := \mathbb{E}[\xi_{k+1} \mid X_k = x].$$

The stationary mean drift is $\mu := \pi^T \bar{\xi}$. The centered drift is $h := \bar{\xi} - \mu\mathbf{1}$.

Theorem 4.1 (Markov additive martingale corrector)**.** *Assume B is ergodic with stationary distribution π, and let h be centered ($\pi^T h = 0$). Let g be the unique centered solution of the Poisson equation*

$$(I - B)g = h, \qquad \pi^T g = 0. \tag{10}$$

Then the compensated process

$$\mathcal{M}_k := Y_k - k\mu + g(X_k) - g(X_0) \tag{11}$$

is a martingale.

Theorem 4.1 is a standard way to construct martingales from additive functionals via Poisson's equation; see, e.g., [10, 11] for general treatments.

The microprice martingale (Theorem 3.14) is the special case $Y_k = p_k$ and $h = G_1 - \mu\mathbf{1}$.

Scope. Theorem 4.1 is purely structural: once an observable is written as an additive functional with conditional mean drift $\bar{\xi}(x)$, solving the Poisson equation (10) produces the drift-corrected martingale. We omit application-specific choices of ξ and focus on the operator construction.

4.3 Operator notation: potentials, projections, and generalized "micro-" transforms

Recasting the Poisson equation in operator language clarifies what the construction really does: the map from drift profiles to correctors is a single linear operator, and all microprice-type constructions are instances of applying it.

It is convenient to phrase the preceding results as statements about linear operators on functions on the state space. Identify a function $f : \mathcal{X} \to \mathbb{R}$ with its coordinate vector in $\mathbb{R}^L$. The post-move chain B then acts as a Markov operator

$$(\mathsf{B}f)(x) := \mathbb{E}[f(X_{k+1}) \mid X_k = x] = \sum_y B_{xy} f(y).$$

Let π be the stationary distribution of B and define the rank-one projection onto constants

$$(\Pi f)(x) := (\pi^T f)\,\mathbf{1}(x), \qquad \text{equivalently} \qquad \Pi = \mathbf{1}\pi^T,$$

so that $\mathrm{Ran}(\Pi) = \mathrm{span}\{\mathbf{1}\}$ and $\ker(\Pi) = \{f : \pi^T f = 0\}$. On the centered subspace $\ker(\Pi)$, the operator $I - \mathsf{B}$ is invertible and its inverse is the *potential* (or *Poisson-solver*) operator

$$\mathsf{U} := (I - \mathsf{B})^{-1} \text{ on } \ker(\Pi), \qquad \text{so that} \qquad \mathsf{U}h = \sum_{n=0}^{\infty} \mathsf{B}^n h \ \text{ for } \pi^T h = 0. \tag{12}$$

In matrix form, U is represented by the *fundamental matrix / group inverse* expression

$$\mathsf{U}h = (I - B + \Pi)^{-1}h, \qquad (\pi^T h = 0),$$

and one may record the deviation-matrix identity

$$(I - B)^{\#} := (I - B + \Pi)^{-1} - \Pi = \sum_{n=0}^{\infty}(B^n - \Pi),$$

where $(I - B)^{\#}$ is the group inverse of $I - B$ [6], and the series representation is the (discrete-time) deviation-matrix identity [7].

Generalized microvalues. Given any integrable additive functional $Y_{k+1} = Y_k + \xi_{k+1}$ with conditional drift $\bar{\xi}(x) = \mathbb{E}[\xi_{k+1} \mid X_k = x]$ and stationary mean $\mu = \pi^T\bar{\xi}$, define the centered drift $h = \bar{\xi} - \mu\mathbf{1}$. Then Theorem 4.1 says that the unique martingale corrector is exactly the potential

$$g = \mathsf{U}h,$$

and the compensated process $Y_k - k\mu + g(X_k) - g(X_0)$ is a martingale. In this sense, the map $h \mapsto \mathsf{U}h$ is the canonical "micro-" transform associated with the Markov skeleton B: it takes a predictable drift profile and returns the unique centered correction that removes it.

The connection between the infinite-horizon corrector and finite-horizon partial sums is made precise by the following identity.

Finite-horizon identity. If h is centered and $g = \mathsf{U}h$ solves the Poisson equation $(I - B)g = h$ with $\pi^T g = 0$, then for every $n \geq 1$,

$$\sum_{\ell=0}^{n-1} B^\ell h \;=\; g - B^n g. \tag{13}$$

Since $B^n g \to \Pi g = 0$ as $n \to \infty$, (13) makes explicit that $g(X_k)$ is the horizon-free limit of the expected *centered* cumulative drift from time k onward.

4.4 Vector-valued functionals

All results extend componentwise to $\mathbb{R}^d$-valued rewards: one solves the same Poisson equation with multiple right-hand sides to obtain a vector martingale corrector.

5 Value functions and Neumann-series resolvents

The constructions of Sections 3–4 work in an undiscounted setting: the Poisson equation requires centering ($\pi^T h = 0$) because the post-move chain B has eigenvalue 1. Introducing a discount or continuation operator Γ with $\|\Gamma M\| < 1$ regularizes the problem: the inverse $(I - \Gamma M)^{-1}$ exists unconditionally, and the solution is a convergent Neumann series. The microprice operator is recovered as the centered limit when the discount vanishes ($\Gamma \uparrow I$). The fixed-point/resolvent viewpoint is standard in dynamic programming and Markov decision processes; see, e.g., [13].

5.1 Neumann-series value functions

Let $M \in \mathbb{R}^{L\times L}$ be a row-stochastic matrix (a Markov kernel on a finite state space $\mathcal{X}$). Let $\Gamma \in \mathbb{R}^{L\times L}$ be a bounded linear operator with

$$\|\Gamma M\|_\infty < 1, \tag{14}$$

where $\|\cdot\|_\infty$ denotes the induced ℓ^∞ operator norm. Given a bounded payoff/reward $u : \mathcal{X} \to \mathbb{R}$, consider the linear fixed-point equation

$$V = u + \Gamma M V. \tag{15}$$

This template covers a wide range of discounted or randomized-horizon objectives; the only property used below is the contraction condition (14).

Proposition 5.1 (Neumann-series value function)**.** *Assume* (14)*. Then* (15) *has a unique bounded solution given by*

$$V = (I - \Gamma M)^{-1} u = \sum_{k=0}^{\infty} (\Gamma M)^k u. \tag{16}$$

Moreover,

$$\|V\|_\infty \le \frac{1}{1 - \|\Gamma M\|_\infty} \|u\|_\infty.$$

5.2 Examples of continuation operators

The template (15) separates the *Markov skeleton* M from the *continuation* mechanism Γ. A few special cases are worth recording because they unify many constructions that appear in the microstructure literature.

Geometric discounting. With $\Gamma = \alpha I$ for $\alpha \in (0, 1)$ one has

$$V = u + \alpha M V, \qquad \text{so} \qquad V(x) = \mathbb{E}_x \left[\sum_{n=0}^{\infty} \alpha^n u(X_n) \right].$$

In this case $V = (I - \alpha M)^{-1} u$ is the usual discounted resolvent, and $\alpha \uparrow 1$ approaches the "undiscounted" boundary.

State-dependent killing / randomized horizons. If $\Gamma = D_\gamma$ is diagonal with $\gamma(x) \in [0, 1)$, then

$$V = u + D_\gamma M V$$

corresponds to a Markov chain that *continues* from state x with probability $\gamma(x)$ and terminates with probability $1 - \gamma(x)$. The contraction condition is simply $\|D_\gamma M\|_\infty \le \|\gamma\|_\infty < 1$. This captures, for example, objectives stopped at a state-dependent random horizon.

Absorbing-chain fundamentals as resolvents. Any substochastic kernel Q can be viewed as a killed Markov kernel. Its fundamental matrix $N = (I - Q)^{-1} = \sum_{n \geq 0} Q^n$ is exactly the value function associated with $\Gamma M = Q$ [5]. The event-time microprice computation in Section 3 is of this type: Q encodes "continue without a move," and absorption ($\rho(Q) < 1$) guarantees the Neumann series converges. In contrast, the post-move chain B has eigenvalue 1, so the undiscounted inverse requires centering.

5.3 Microprice as a centered boundary case

For $\alpha \in (0, 1)$, the discounted potential $(I - \alpha B)^{-1}$ is well-defined on all of $\mathbb{R}^L$ and admits the Neumann series

$$(I - \alpha B)^{-1} v = \sum_{k=0}^{\infty} \alpha^k B^k v.$$

As $\alpha \uparrow 1$, the resolvent becomes singular because B has eigenvalue 1. On the centered subspace $\{v : \pi^T v = 0\}$, the limit exists and equals the microprice operator.

Proposition 5.2 (Resolvent limit)**.** *On the centered subspace $\{v : \pi^T v = 0\}$, the discounted resolvent $(I - \alpha B)^{-1}$ converges as $\alpha \uparrow 1$ to the microprice operator $(I - B + \mathbf{1}\pi^T)^{-1}$.*

6 Discussion and limitations

The literature often uses "microprice" both for the instantaneous size-weighted mid m_t and for the long-horizon fair value $p_k + G^*(X_k)$; compare [1] and the terminology discussion in the LOB survey [4]. Keeping these distinct avoids conceptual confusion.

The condition $\pi^T G_1 = 0$ asserts zero stationary drift per move and is substantive. If the drift is nonzero (trends, regime shifts), the undiscounted microprice is ill-posed; one should work with centered Poisson equations or with well-posed resolvents satisfying a Neumann-series condition (Section 5).

Bounds based on $\|N\|_\infty \leq 1/\varepsilon$ require a uniform lower bound on move probabilities. In practice, some states may exhibit extremely rare moves, making N ill-conditioned and amplifying estimation error.

The framework is operator-theoretic: it applies to whatever finite-state Markov approximation is used. If the chosen state is not approximately Markov, the resulting kernels are unstable and predictive performance degrades; this is an empirical modelling issue rather than a limitation of the algebra.

7 Conclusion

Stoikov's microprice can be read as an operator identity rather than a collection of model variants. At mid-price move times, the mid-price is a Markov additive process; the microprice adjustment is the Poisson corrector on the post-move chain. Adding this corrector to the price produces a martingale, and the same Poisson equation constructs martingales for general additive order-book functionals.

A second message is that many "fair value" constructions are value functions for the same Markov skeleton. When the relevant operator inverse satisfies a Neumann-series condition, the value function is a well-posed resolvent. The microprice corresponds to the boundary case

where the eigenvalue at 1 forces centering; the microprice operator is the centered limit of discounted resolvents.

Several directions remain open. On the modelling side, principled state-space selection (minimizing truncation error or maximizing Markov fidelity) deserves a systematic treatment. In nonstationary markets the centering condition $\pi^T G_1 = 0$ may fail; dynamic re-centering or regime-adaptive kernels are natural extensions. Finally, a continuous-time formulation via CTMC generators and their resolvents would connect the discrete framework directly to the continuous-time microstructure literature.

A Proofs

Proof of Lemma 3.2. For any matrix A, the spectral radius satisfies $\rho(A) \leq \|A\|$ in any submultiplicative operator norm. Taking the induced ℓ^∞ norm,

$$\|Q\|_\infty = \max_x \sum_y |Q_{xy}| = \max_x \sum_y Q_{xy} \leq 1 - \varepsilon.$$

Hence $\rho(Q) \leq 1 - \varepsilon < 1$. □

Proof of Proposition 3.3. Since $\rho(Q) < 1$, the Neumann series $\sum_{k=0}^\infty Q^k$ converges absolutely and equals $(I - Q)^{-1}$. For the norm bound,

$$\|N\|_\infty \leq \sum_{k=0}^\infty \|Q^k\|_\infty \leq \sum_{k=0}^\infty \|Q\|_\infty^k \leq \sum_{k=0}^\infty (1-\varepsilon)^k = \frac{1}{\varepsilon}.$$

The probabilistic interpretation is standard for absorbing chains: N_{xy} is the expected number of visits to y before absorption when starting from x; see, e.g., [5]. □

Proof of Lemma 3.5. From $N = I + Q + Q^2 + \cdots$ and non-negativity of Q, each term is non-negative entrywise. Thus $N \geq 0$ and $N_{xx} \geq 1$ for all x. □

Proof of Lemma 3.7. Let τ be the first event-time index at which a mid-price move occurs, as defined in Lemma 3.7. Conditioning on the first step from x gives the first-step equation

$$G_1(x) = \sum_y Q_{xy} G_1(y) + \sum_y T_{xy}\, \mathbb{E}[\Delta p \mid x \to y,\ \text{move}].$$

By definition, the second term is $\bar{r}_x$, hence $G_1 = \bar{r} + QG_1$ and $(I - Q)G_1 = \bar{r}$. Since $\rho(Q) < 1$, $I - Q$ is invertible and $G_1 = (I - Q)^{-1}\bar{r}$.

For the embedded chain, note that

$$\mathbb{P}_x(X_\tau = y) = \sum_{k=0}^\infty \mathbb{P}_x(X_k = y,\ \tau = k+1) = \sum_{k=0}^\infty (Q^k T)_{xy} = \big((I-Q)^{-1}T\big)_{xy} = B_{xy}.$$

□

Proof of Lemma 3.8. The move times (τ_k) are stopping times for the event-time filtration, and the event-time chain (X_n) is Markov with kernel $P = Q + T$. By the strong Markov property, the post-τ_k evolution of (X_n) depends on the past only through X_{τ_k}. The transition probabilities of the sampled chain follow:

$$\mathbb{P}(X_{\tau_{k+1}} = y \mid X_{\tau_k} = x) = \sum_{n=1}^\infty \mathbb{P}(X_{\tau_k+n} = y,\ \tau_{k+1} = \tau_k + n \mid X_{\tau_k} = x) = \sum_{n=0}^\infty (Q^n T)_{xy} = B_{xy},$$

where the last equality is the calculation from Lemma 3.7. □

Proof of Lemma 3.9. From $(Q + T)\mathbf{1} = \mathbf{1}$, we have $T\mathbf{1} = (I - Q)\mathbf{1}$. Thus

$$B\mathbf{1} = (I - Q)^{-1}T\mathbf{1} = (I - Q)^{-1}(I - Q)\mathbf{1} = \mathbf{1}.$$

Non-negativity of B follows from non-negativity of N and T. □

Proof of Proposition 3.16. By the embedded-chain interpretation (Lemma 3.7), $G_1(x) = \mathbb{E}_x[\Delta p_\tau]$. Since $|\Delta p_\tau| \le D$ almost surely (Assumption 2), Jensen's inequality gives $|G_1(x)| \le \mathbb{E}_x[|\Delta p_\tau|] \le D$. Hence $\|G_1\|_\infty \le D$. For the operator-norm bound, $\|G_1\|_\infty = \|N\bar{r}\|_\infty \le \|N\|_\infty \|\bar{r}\|_\infty \le \|\bar{r}\|_\infty/\varepsilon$. □

Proof of Theorem 3.13. Write

$$B = \mathbf{1}\pi^T + C, \qquad C := B - \mathbf{1}\pi^T.$$

Since $B\mathbf{1} = \mathbf{1}$ and $\pi^T B = \pi^T$, we have $C\mathbf{1} = 0$ and $\pi^T C = 0$. Ergodicity implies that 1 is a simple eigenvalue of B and all other eigenvalues have modulus < 1, so $\rho(C) = \lambda_* < 1$.

Centering gives $\mathbf{1}\pi^T G_1 = 0$, hence for all $k \ge 0$,

$$B^k G_1 = (\mathbf{1}\pi^T + C)^k G_1 = C^k G_1.$$

Therefore,

$$\sum_{k=0}^{\infty} B^k G_1 = \sum_{k=0}^{\infty} C^k G_1 = (I - C)^{-1} G_1,$$

which proves the rank-one inverse formula since $I - C = I - B + \mathbf{1}\pi^T$. For centering, note that $\pi^T(I - B + \mathbf{1}\pi^T) = \pi^T$, so $\pi^T(I - B + \mathbf{1}\pi^T)^{-1} = \pi^T$, and therefore $\pi^T G^* = \pi^T G_1 = 0$.

To verify the Poisson equation, start from the partial sums $S_n := \sum_{k=0}^{n-1} B^k G_1$. Then

$$(I - B)S_n = G_1 - B^n G_1.$$

Since $B^n G_1 = C^n G_1 \to 0$ as $n \to \infty$ (because $\rho(C) < 1$), taking $n \to \infty$ yields $(I-B)G^* = G_1$. Uniqueness under the centering condition follows because the nullspace of $I - B$ is $\mathrm{span}\{\mathbf{1}\}$: if $(I - B)g = 0$ and $\pi^T g = 0$, then g must be constant and hence $g \equiv 0$.

For truncation, note that $G^* - S_n = \sum_{k=n}^{\infty} C^k G_1$. In finite dimensions, for any $\rho \in (\rho(C), 1)$ there exists $C_\rho < \infty$ such that $\|C^k\|_\infty \le C_\rho \rho^k$ for all k. Hence

$$\|G^* - S_n\|_\infty \le \sum_{k=n}^{\infty} \|C^k\|_\infty \, \|G_1\|_\infty \le \sum_{k=n}^{\infty} C_\rho \rho^k \|G_1\|_\infty = \frac{C_\rho \rho^n}{1 - \rho} \|G_1\|_\infty.$$

□

Proof of Theorem 3.14. Using the Markov property at move times,

$$\begin{aligned}
\mathbb{E}[\mathcal{P}_{k+1} \mid \mathcal{F}_k] &= \mathbb{E}[p_{k+1} + G^*(X_{k+1}) \mid \mathcal{F}_k] \\
&= p_k + \mathbb{E}[\Delta p_{k+1} \mid X_k] + \mathbb{E}[G^*(X_{k+1}) \mid X_k] \\
&= p_k + G_1(X_k) + (BG^*)(X_k).
\end{aligned}$$

From $(I-B)G^* = G_1$ we have $G_1 + BG^* = G^*$ pointwise, hence $\mathbb{E}[\mathcal{P}_{k+1} \mid \mathcal{F}_k] = p_k + G^*(X_k) = \mathcal{P}_k$. □

Proof of Theorem 4.1. Let $\mathcal{M}_k$ be defined by (11). Then

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \xi_{k+1} - \mu + g(X_{k+1}) - g(X_k).$$

Taking conditional expectation given $\mathcal{F}_k$ and using $\mathbb{E}[\xi_{k+1} \mid X_k] = \bar{\xi}(X_k)$ and $\mathbb{E}[g(X_{k+1}) \mid X_k] = Bg(X_k)$ gives

$$\mathbb{E}[\mathcal{M}_{k+1} - \mathcal{M}_k \mid \mathcal{F}_k] = (\bar{\xi} - \mu\mathbf{1})(X_k) + (Bg - g)(X_k) = h(X_k) - (g - Bg)(X_k) = 0,$$

where the last equality is (10) written as $g - Bg = h$. Hence $(\mathcal{M}_k)$ is a martingale. □

Proof of Proposition 5.1. Assume $\|\Gamma M\|_\infty < 1$. Then the Neumann series $\sum_{k=0}^{\infty}(\Gamma M)^k$ converges absolutely in the induced operator norm, and defines a bounded linear operator

$$S := \sum_{k=0}^{\infty}(\Gamma M)^k.$$

Telescoping gives

$$(I - \Gamma M)S = \sum_{k=0}^{\infty}(\Gamma M)^k - \sum_{k=1}^{\infty}(\Gamma M)^k = I,$$

so $S = (I - \Gamma M)^{-1}$. Setting $V = Su$ yields (16), and multiplying by $(I - \Gamma M)$ gives the fixed point equation (15).

For uniqueness, if V and W both solve (15), then $V - W = \Gamma M(V - W)$. Taking $\|\cdot\|_\infty$ norms gives $\|V - W\|_\infty \le \|\Gamma M\|_\infty \|V - W\|_\infty$, hence $V = W$.

Finally,

$$\|V\|_\infty \le \sum_{k=0}^{\infty} \|\Gamma M\|_\infty^k \, \|u\|_\infty = \frac{1}{1 - \|\Gamma M\|_\infty} \, \|u\|_\infty.$$

□

Proof of Proposition 5.2. Let v satisfy $\pi^T v = 0$ and write $B = \mathbf{1}\pi^T + C$ with $C = B - \mathbf{1}\pi^T$ as above. Then $B^k v = C^k v$ for all $k \ge 0$. For $\alpha \in (0, 1)$,

$$(I - \alpha B)^{-1} v = \sum_{k=0}^{\infty} \alpha^k B^k v = \sum_{k=0}^{\infty} \alpha^k C^k v = (I - \alpha C)^{-1} v.$$

Since $\rho(C) < 1$, we have $(I - \alpha C)^{-1} \to (I - C)^{-1}$ as $\alpha \uparrow 1$. Finally, $I - C = I - B + \mathbf{1}\pi^T$, so the limit is $(I - B + \mathbf{1}\pi^T)^{-1} v$. □

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